# How Effective are Large Language Models for Sports Betting?

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# Some Background

What is AI and Large Language Models?

What is a Money Line Bet?

Basics of Sports Betting

# How are Payouts Calculated?

**Odds can be positive or Negative**

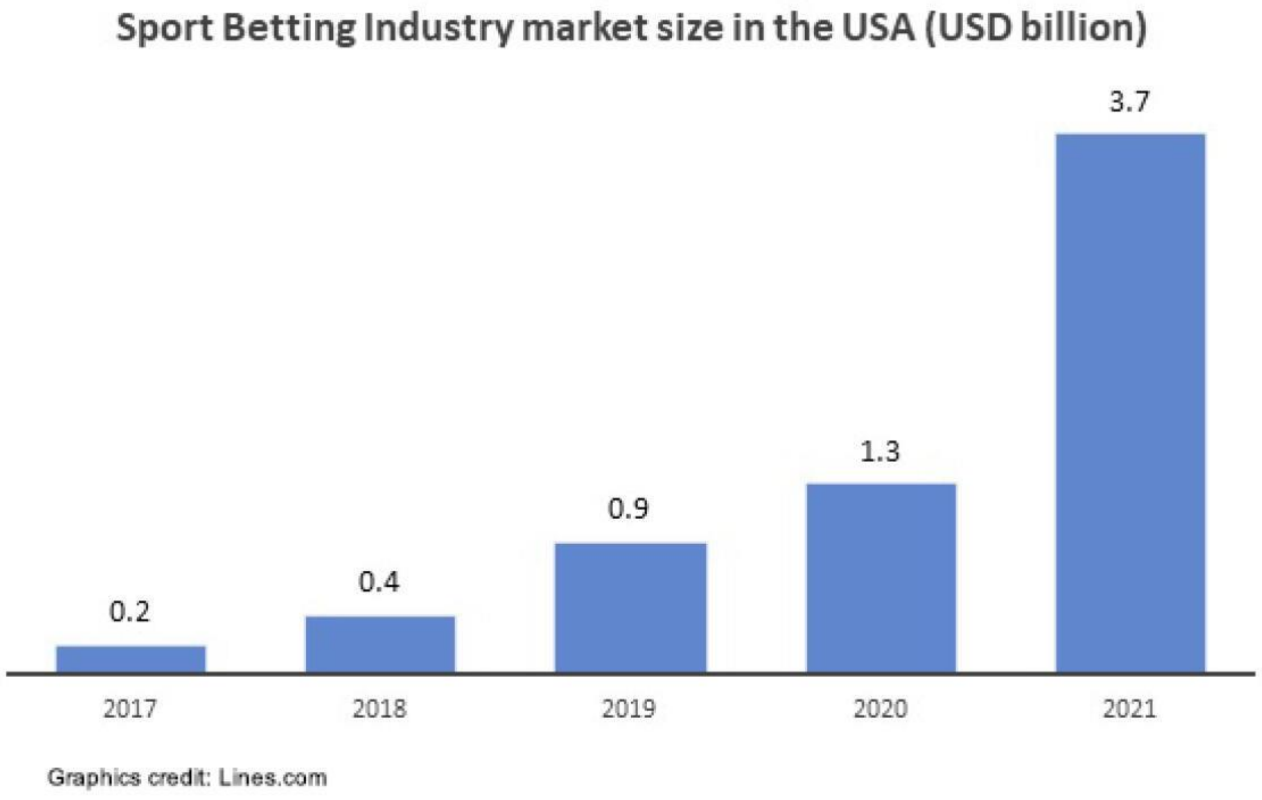
- Positive = Bet Amount * (Odds / 100)
- Negative = Bet Amount / (Odds / -100)

**Positive Odds = Underdog won**

**Negative Odds = Favorite won**

# Why is this Important?

- Sports Betting has become a billion-dollar industry
- Impact of COVID
- Recent Development = Best Opportunity
- **How to start?**

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**Objective:** Determine which of Three Large Language models is most effective for NBA Money Line Betting

# Project Details

- 3 LLMs used
  - Meta, Gemini and Copilot (latest versions as of April 1st, 2025)
- Each LLM had 3 Inputs Types
  - No data, Data, Upset
- Input was the same, and recorded in a doc
- Data was kept in an Excel sheet
  - Both can be accessed here: https://github.com/KonnorBoucher/Senior_Research_Project
- DraftKings used as bookkeeper for odds, Odds Here: (https://sportsbook.draftkings.com/leagues/basketball/nba?wpcid=605461783&wpsnetn=o&wpkwn=draftkings%20nba%20odds&wpkmatch=e&wpcrid=&wpscid=1334809637221892&wpkwid=kwd-83426762688010:loc-4086&an=adwords&gclid=68ce10cc11ab18c6e1d778d03564d457&gclsrc=3p.ds&msclkid=68ce10cc11ab18c6e1d778d03564d457&utm_source=bing&utm_medium=cpc&utm_campaign=SB_NAT_Brand%2B_Sports&utm_term=draftkings%20nba%20odds&utm_content=Brand%2B_All%20Devices_Basketball)
- 115 Total games, 15 days
- Assumed bet for each = $5

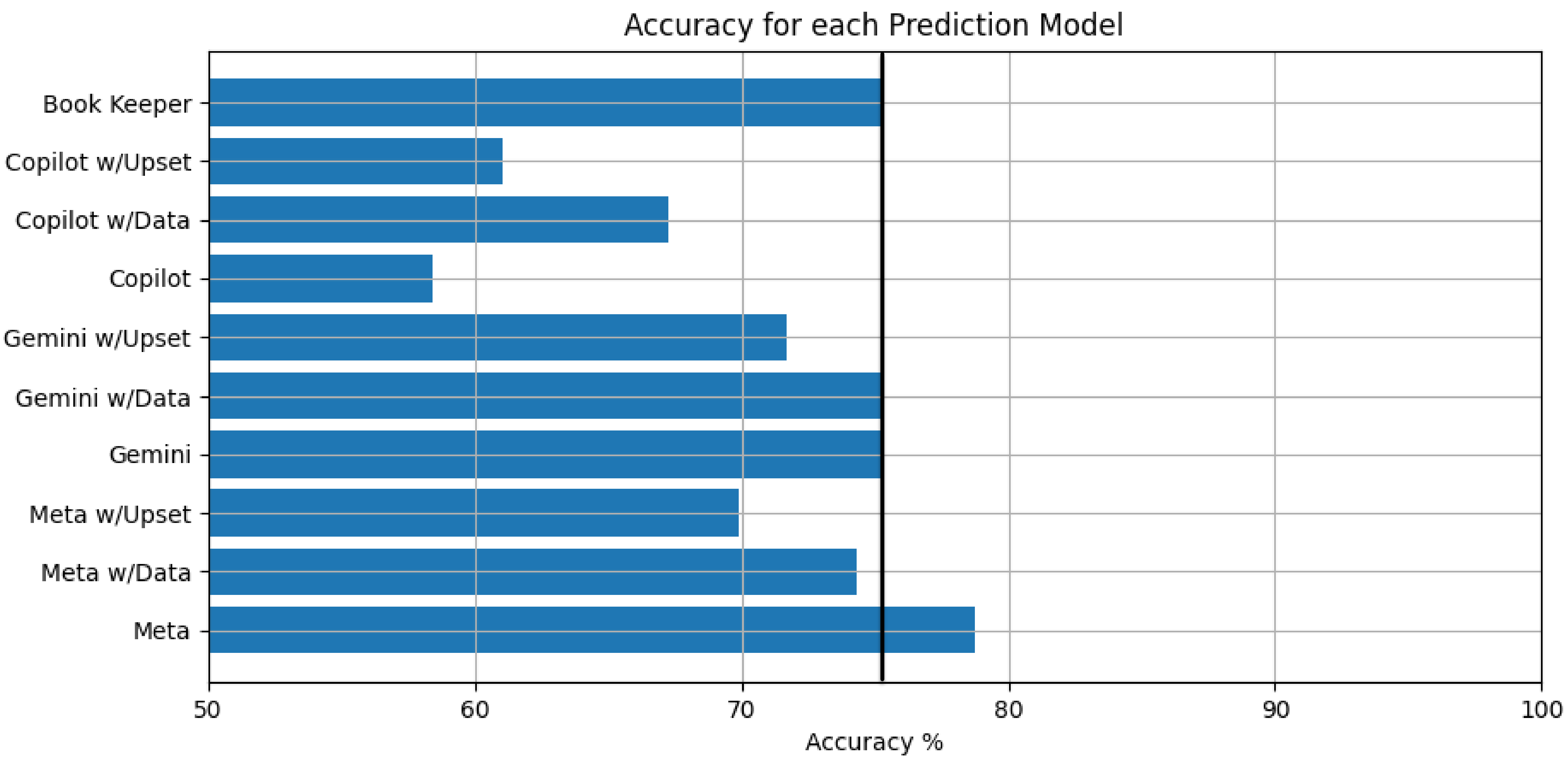
# Helper Python Program

- **Did the following:**
  - Used pandas to manipulate .csv file
  - Calculated profits
  - Averaged out values
  - Created 6 unique figures, (4 shown)

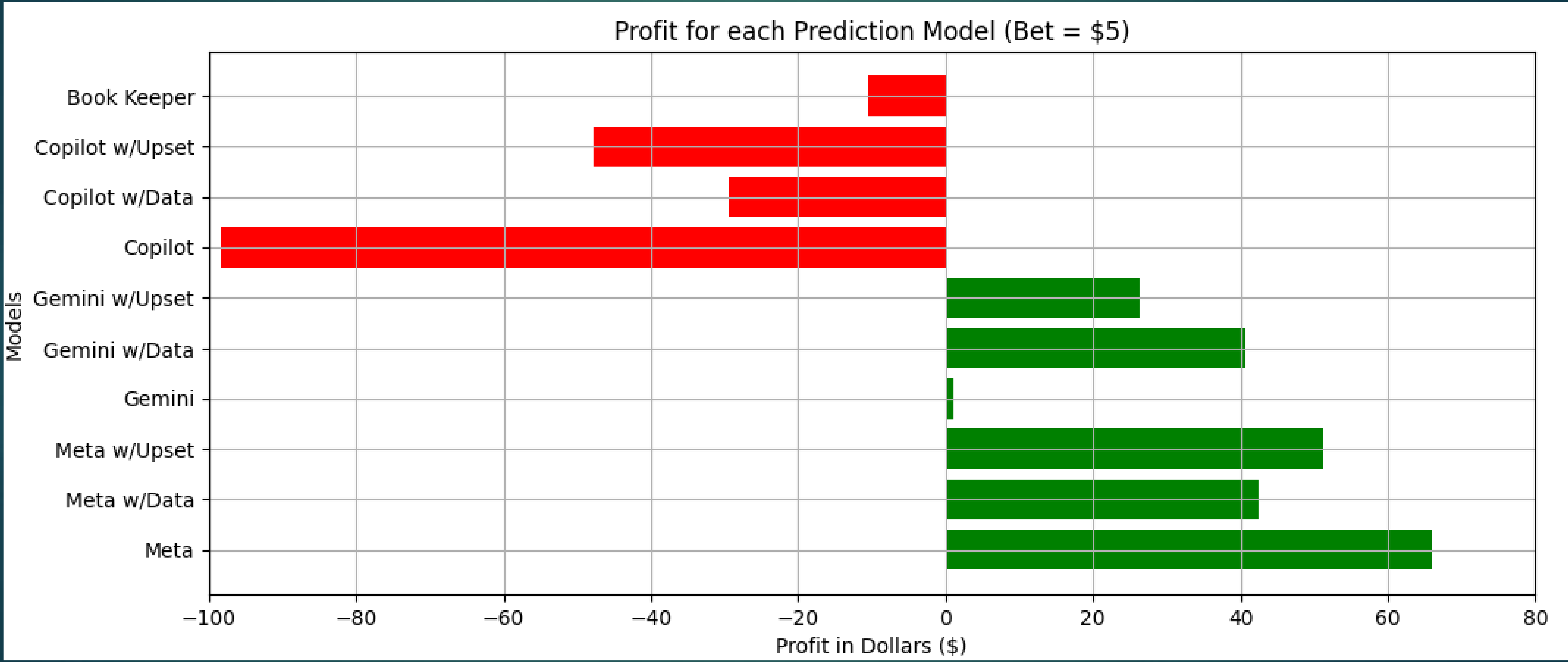
# Example Part of Program

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

def calc_prof(bet_amt, odds, correct):
    if correct == False:
        return bet_amt * -1
    else:
        if odds < 0:
            return bet_amt / (odds / -100)
        else:
            return bet_amt * (odds / 100)

df = pd.read_csv('Basketball AI Data.csv')

total_rows = df.shape[0]
pd.options.display.max_rows = total_rows

df = df.drop(df.iloc[:, 14:23], axis = 1)
```

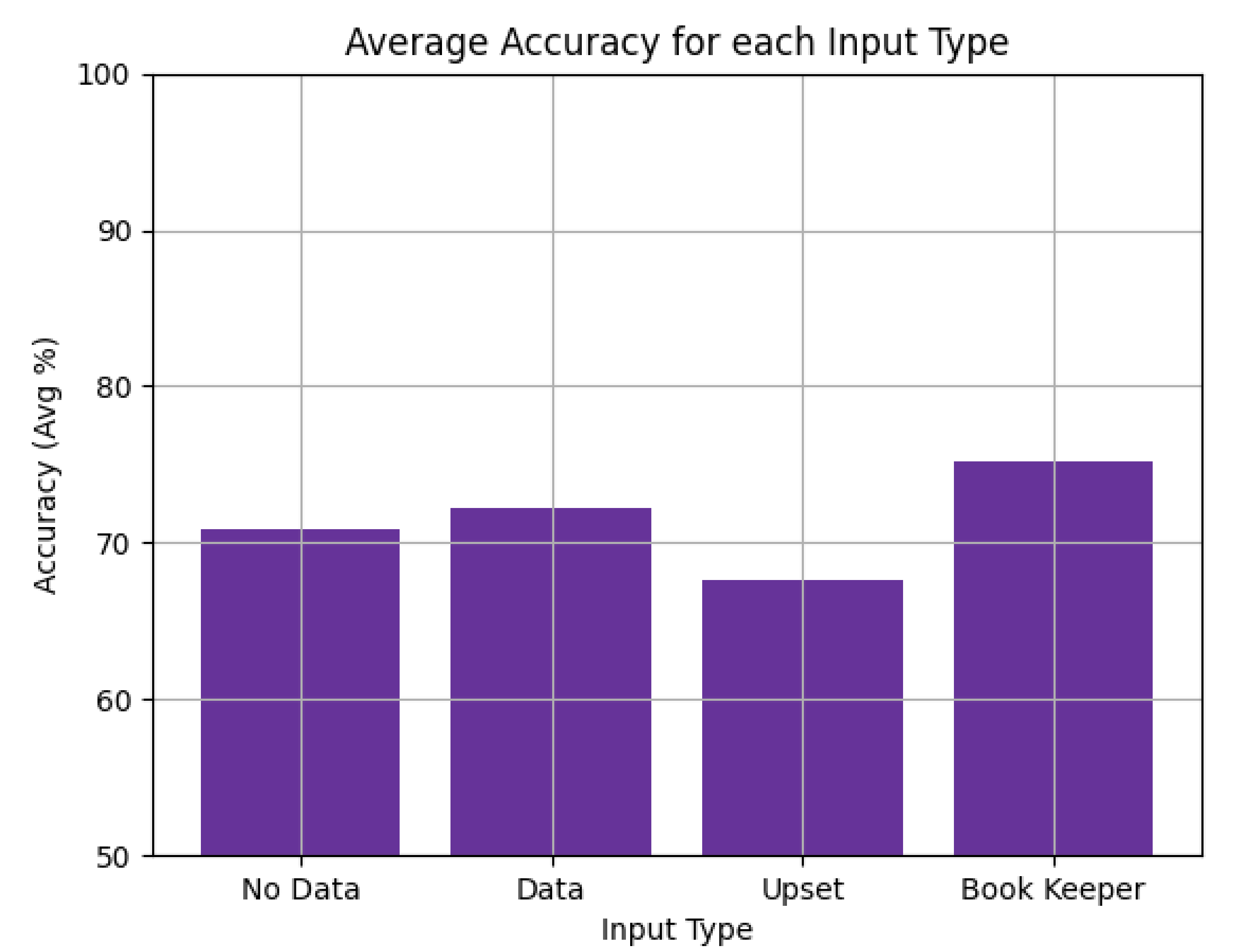
# Accuracy for each Prediction Model



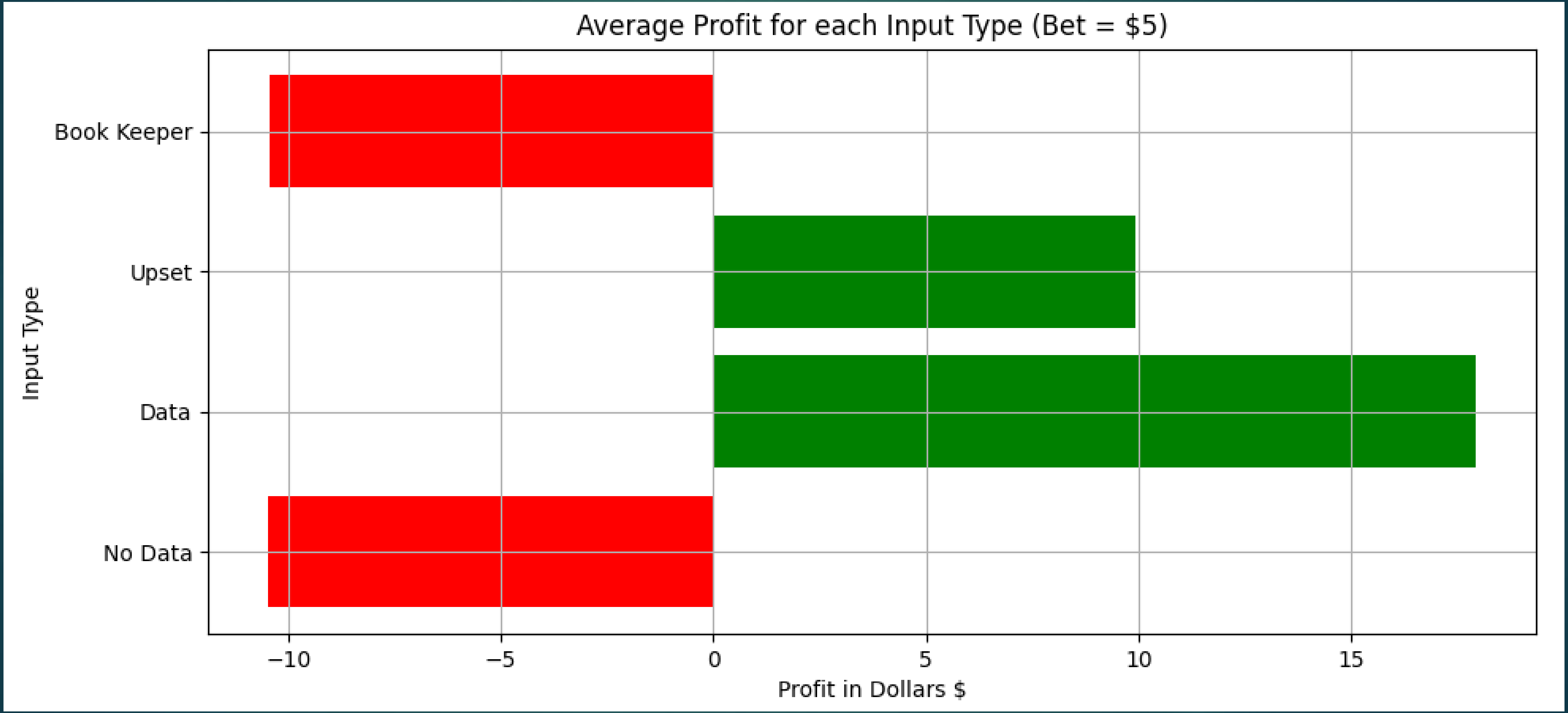
# Profit for each Prediction Model



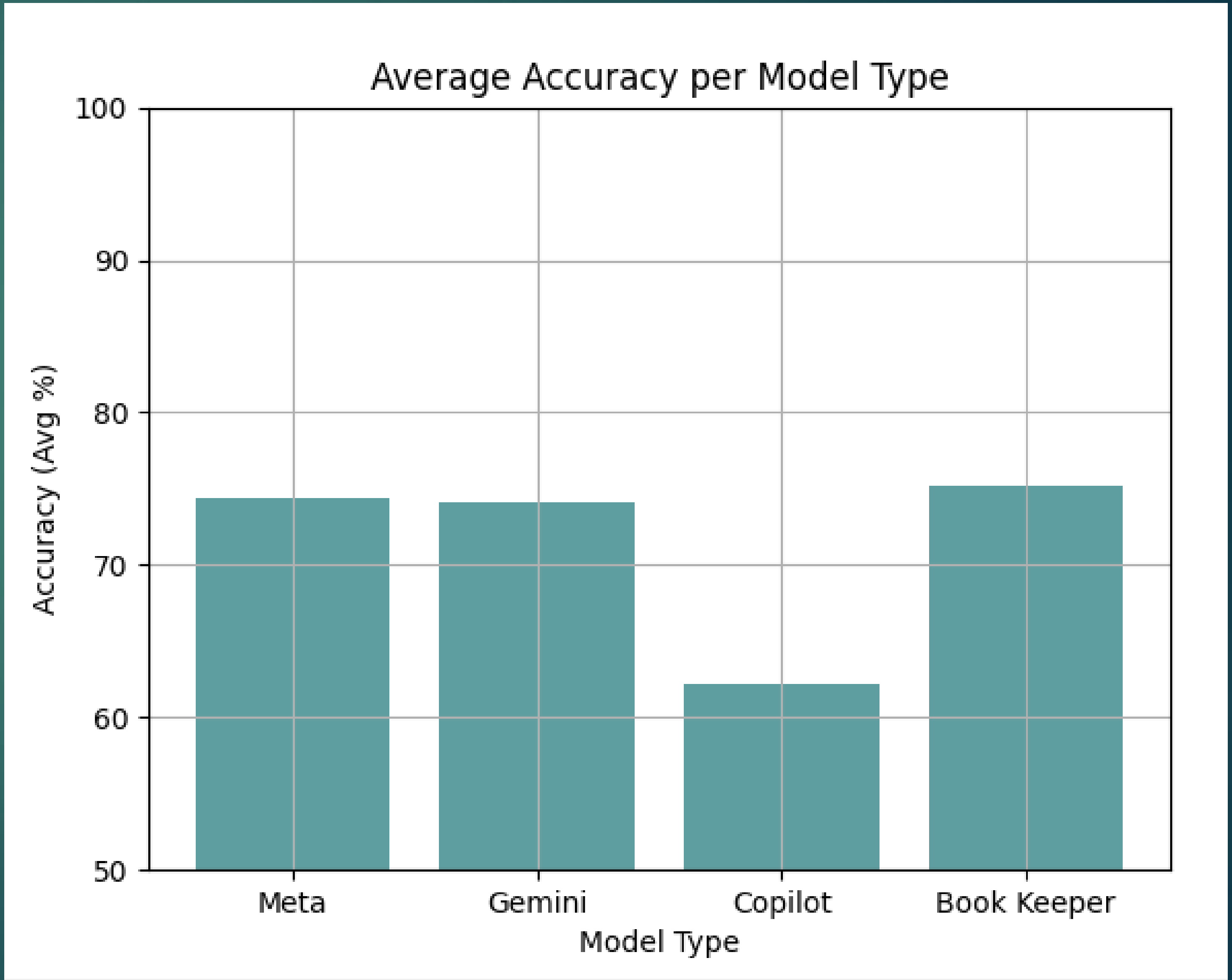
# Average Accuracy per Input Type



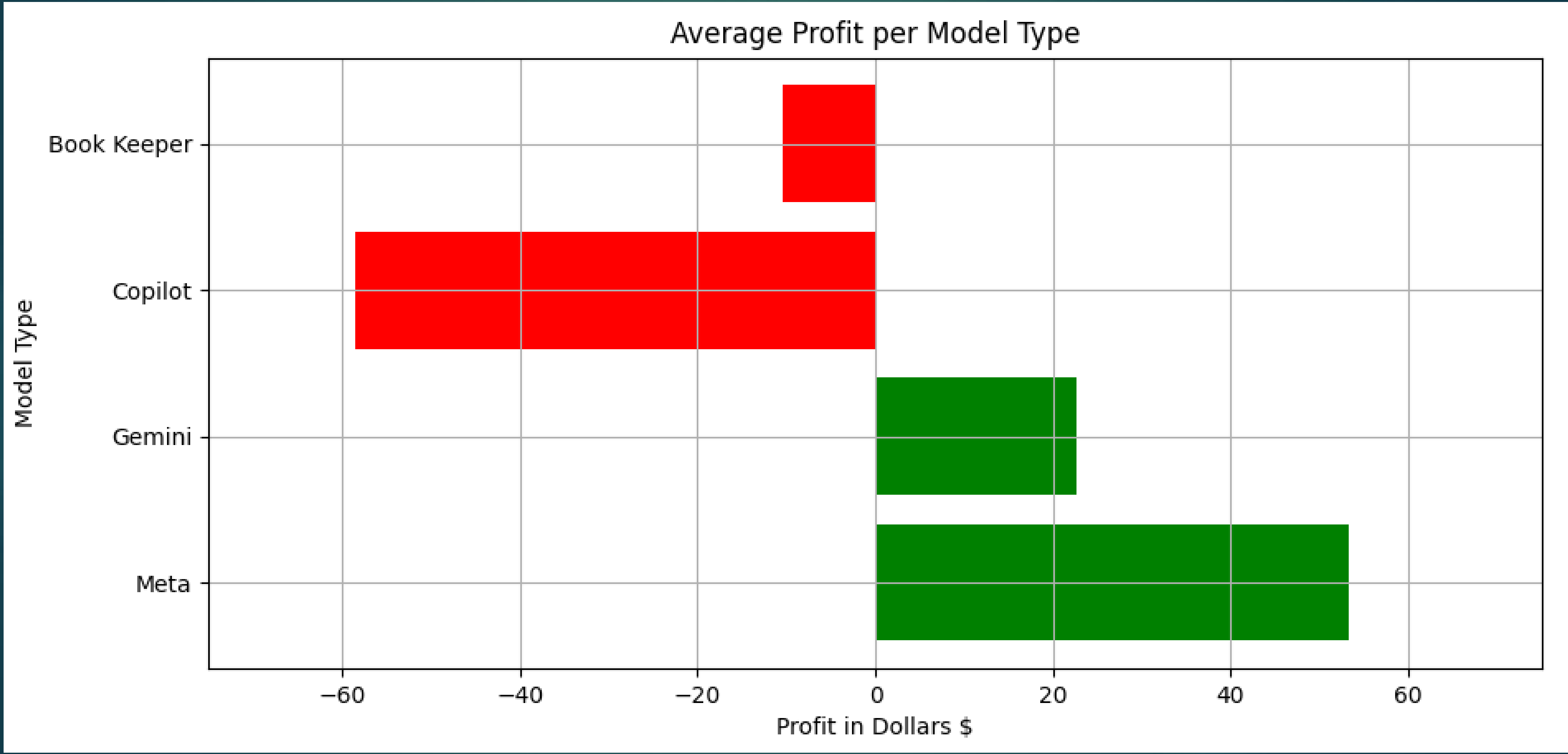
# Average Profit per Input Type



# Average Accuracy per Model



# Average Profit Per Model



# Conclusions

What was the best model?

Any trends?

What can be learned?

# Limitations / Further Study

- Amount of games
- Only NBA Moneyline
- Only 15-day span